IBM Fusion

Regional Disaster Recovery Lab Guide

Matt LeVan

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IBM Fusion: TechZone Quick Start

Introduction

IBM Technology Zone (TechZone) is the single destination for our go-to-market teams and IBM business partners ecosystem to access on-demand and live environments to learn, build, show, and share the value of IBM solutions. Additionally, they can extend our certified base images and further customize them for test, education, demonstration, and pilot activities.

This lab guide is a Quick Start for all Fusion labs on TechZone. The lab guide will explain how to find Fusion content, as well as how to reserve and use the Fusion related lab instances. Lab exercises and specific preparation steps are detailed in the respective lab guides by topic of interest.

This guide can be used to bring a TechZone instance to a general starting point. From this starting point you may continue to the relevant lab guide document or, if you wish, create your own specific configuration for a user defined test or demo.

All TechZone instances for Fusion are based on configurable OpenShift clusters with a virtual machine (VM) node infrastructure deployed on OpenShift Virtualization (OCP-Virt) technology. The underlying VMs have nested virtualization enabled so that they can host a second level VM for demonstrating technologies such as OCP-Virt and Hosted Control Planes (HCP aka, "Control Tower").

Product disclaimer

IBM Fusion is being developed and released in an agile manner. In addition to adding new capabilities, the interface is likely to change over time. Therefore, the screenshots used in this document and other lab guides may not always look exactly like what you see in the product. Depending timing, you can expect to encounter some of the following:

- Changes in the user interface (UI), such as the location of buttons or text in various fields
- Additional tabs or buttons

These differences should not affect how the lab exercises work. When changes are observed, have patience and explore.

You will need an IBM account and corresponding IBMid to gain access to IBM Technology Zone.

Getting help

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If you require assistance in interpreting any of the steps in this demo, please post your questions to the #storage_demo_feedback Slack channel for IBMers only. Business Partners can request help at the Partner Plus Support website.

If you have an issue with TechZone or want to open a support case go to the IBM Technology Zone Help. Alternatively, you can send an email to techzone.help@ibm.com. This contact information is also provided in the emails that you receive from IBM Technology Zone.

For help with the Fusion product see the Fusion documentation.

Prerequisites & getting started

You **must** be familiar with the Linux operating system, as well as know how to use the command line interface to execute basic Linux commands such as **kubectl**, **oc**, **exit**, **ls**, **mkdir**, **ssh**, **sudo**, and **curl** (to name few). Ideally, you should also have some experience working with the Red Hat OpenShift Graphical User Interface (GUI) and executing Red Hat OpenShift Container Platform and Kubernetes command line interface (CLI) commands. If you need experience using Linux, Kubernetes and Red Hat OpenShift Container Platform, it is highly recommended that you complete the following Red Hat courses *before* you attempt to work through the exercises in this lab.

Red Hat Courses:

- DO080 Containers Kubernetes and Red Hat OpenShift Technical Overview
- Red Hat OpenShift: Technical Overview

Note Red Hat Partner Connect Training access required.

When instructions are given to enter text, the text is usually shown in mono space font `this is an example`. This is to help indicate that you need to type or copy/paste all the text as is.

IBM Fusion on TechZone

IBM TechZone can be found at this URL https://techzone.ibm.com/. To access IBM TechZone you must have an IBM ID which you can obtain from your IBM representative as a client, Business Partner or employee.

Once you are logged in to IBM TechZone head over to the IBM Fusion Collection and explore the Environement and Resources.

Bookmark these pages for future use or mark them as Favorites by clicking the heart-shaped icon on the

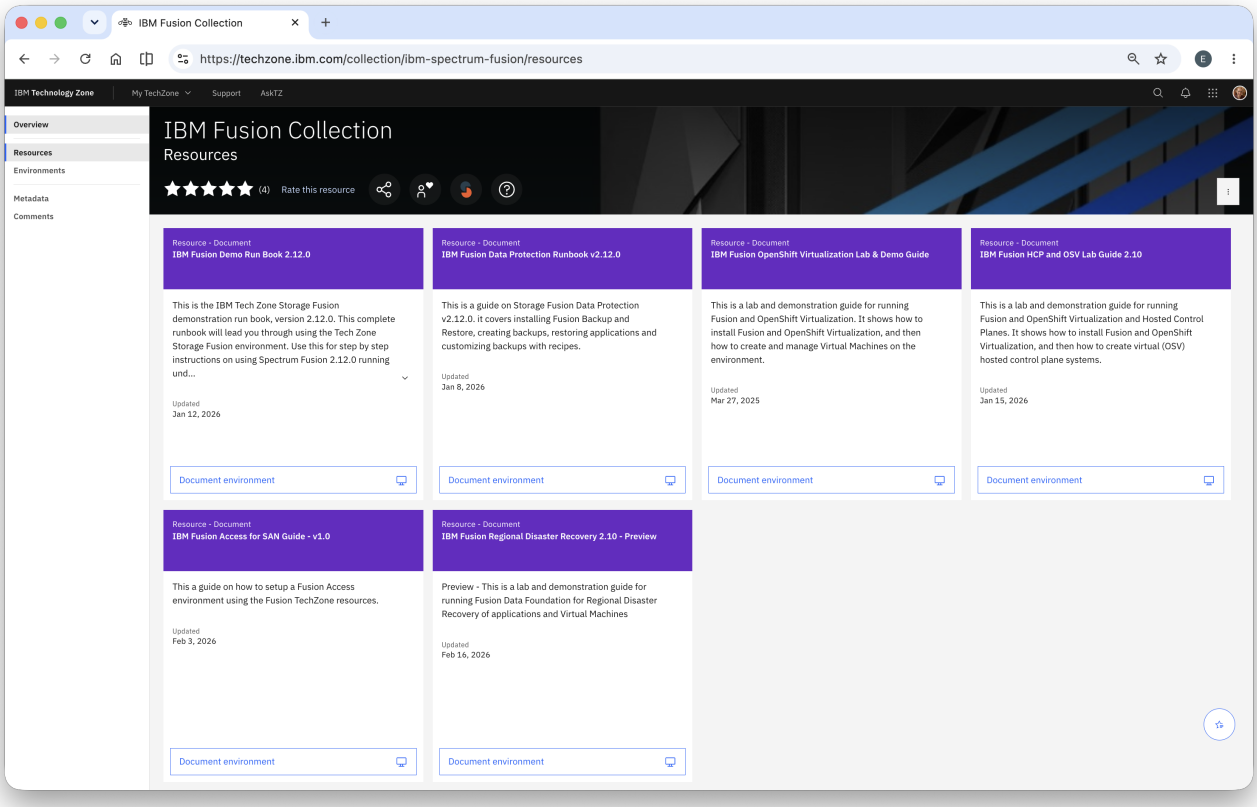
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page.

Resources

The resource page includes links to the available lab guides. These guides are available in PDF format and over time will be also available as on-line viewable material. Due to the rapid development cycles of some products, the guides may not always reflect the latest UI changes and you can expect to encounter some differences. These differences should not affect how the lab exercises work; be patience and explore to enhance your learning.

The lab guides are typically a step by step procedure to performing and learning a set of features and product capabilities. Feel free to explore and experiment!



Environments

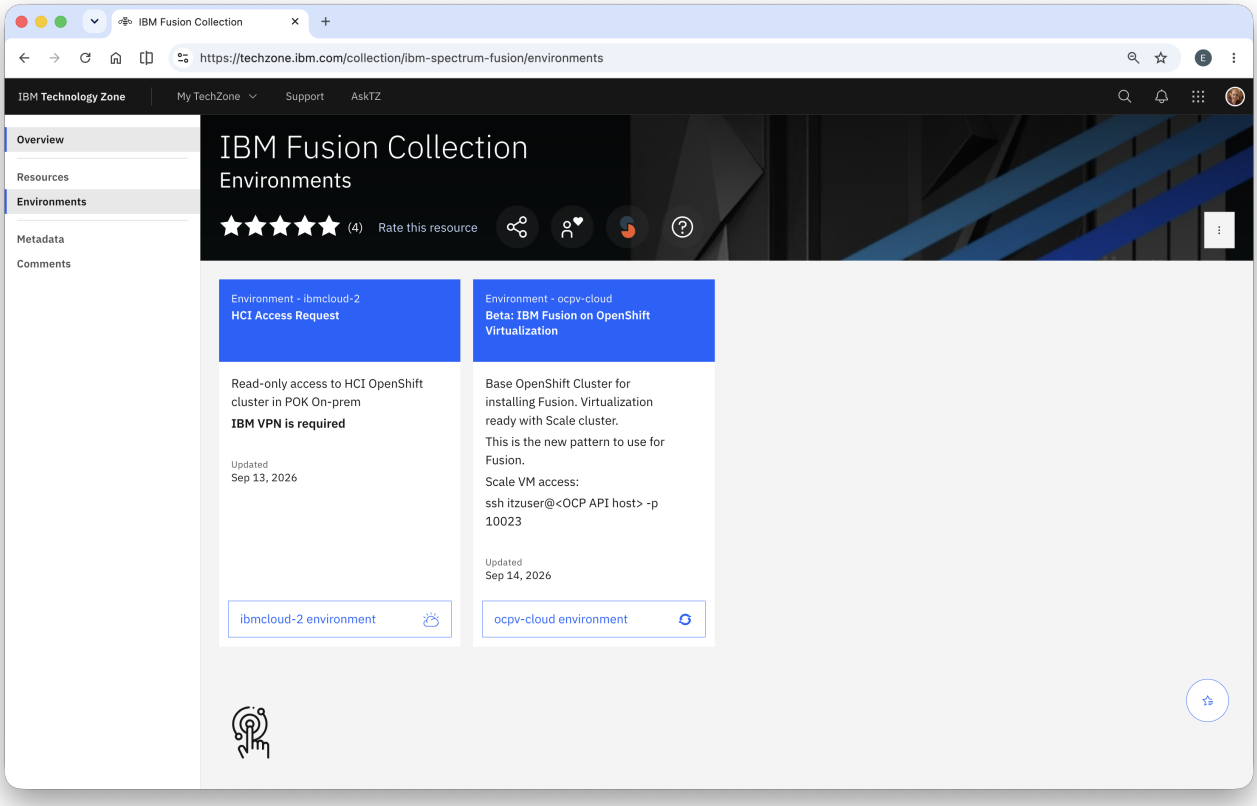
The environments page includes links to available lab resources that you can request for your project. Once your request is provisioned and ready, you will be able to use these resources for your own education, experimentation, and client demos.

The environment that is provisioned for you is an OpenShift (OCP) cluster consisting of your choice of

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components with some extra resources such as a Bastion node and an IBM Storage Scale node. You can use these for the guided labs or use them as a blank slate for your own explorations.

The next sections will walk you through a provisioning and basic use of an OCP cluster on TechZone.



Provisioning a lab

Begin your reservation process by logging in to TechZone and navigating the the Fusion Environments page.

The environments page lists the environment types you can provision and gain access to for your project.

Select the environment titled **Beta: IBM Fusion on OpenShift Virtualization** by clicking Reserve at the bottom of the tile. This will start a provisioning wizard that will walk you through the provisioning process:

1. General

Name your environment with a descriptive name that will help you identify the setup. This name will be visible to all authorized users, including team members, stakeholders, and clients.

Enter a free text description of the environment. Anyone you share this request with will see this

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description when getting access. Include relevant details to help them understand the purpose and expectations.

2. Business purpose

Describe the usage intent, including the context and purpose behind the request. This information helps the TechZone understand the business need and ensure the request is appropriately supported and prioritized.

By choosing a Purpose tile you can indicate the intended use. Each selection has a different reservation time schedule and different requirements for providing an ISC record reference. If you need to provide an ISC record reference simply copy and paste the 18-character Opportunity Number and make sure that the product list in the opportunity include the product name of the lab you are reserving; in our case Fusion.

3. Deployment

TechZone resource reservations rely on IBM Cloud. Choose a location closest to where the session will be hosted (if in person) to optimize performance. While this preference will be considered, deployment to the selected geography is subject to resource availability and cannot be guaranteed.

4. Scheduling

The scheduling page gives you the ability the set the preferred reservation date range within the Policy period determined by your intended use selection you made in the Business purpose page. Provide details on scheduling requirements and the planned dates for the intended request.

Important	Do not finalize dates with clients until the request is approved or until the resources are fully deployed and ready. Always reserve a few days in advance of any meetings or demos.

5. Configuration

Enter clear and detailed configuration choices for the environment request. You can choose to keep the defaults as displayed, or tailor the setup to your needs by clicking Customize and then modifying the environment attributes and then clicking Save.

6. Review and submit

Review the details on this pager before submitting your reservation request. This final step ensures that all required information is accurate and that you are aware of the responsibilities associated with the use of

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IBM TechZone resources.

You can go back and change reservation inputs by clicking the Edit links on the page or the Back button to step backwards in the wizard.

Any wizard step with a blue checkmark icon next to it is properly entered, and any red exclamation point icon should be revisited and corrected before submission.

Once you submit your request the system automatically start a background process to reserve the resources needed for configuring the OpenShift cluster and other resources.

You can track the reservation requests and their status on the My requests page. This page also has a display and sort filter that you can use the manage what you see on the page.

Tip Clear the filter or add the selection of Failed reservations so that you will be able to track reservations that failed to provision properly. This will prevent them from "disappearing" from your view.

You will receive emails from IBM Technology Zone <noreply@techzone.ibm.com> indicating the status of your reservation. The status may indicate a reservation failure or success, and over the duration of use you may also receive reminders for imminent expiration or confirmation of your initiated deletion.

If you receive and expiration notice and you have a date extension option, then follow up and extend your reservation in order to keep using it without disruption.

My reservation failed, what to do?

Reservations may fail from time to time. This depends of many factors including current workloads and available resources. Always reserve in advance of any meetings or demos! if a reservation fails redo the reservation entry using the reservation wizard and described in this section.

Managing a lab

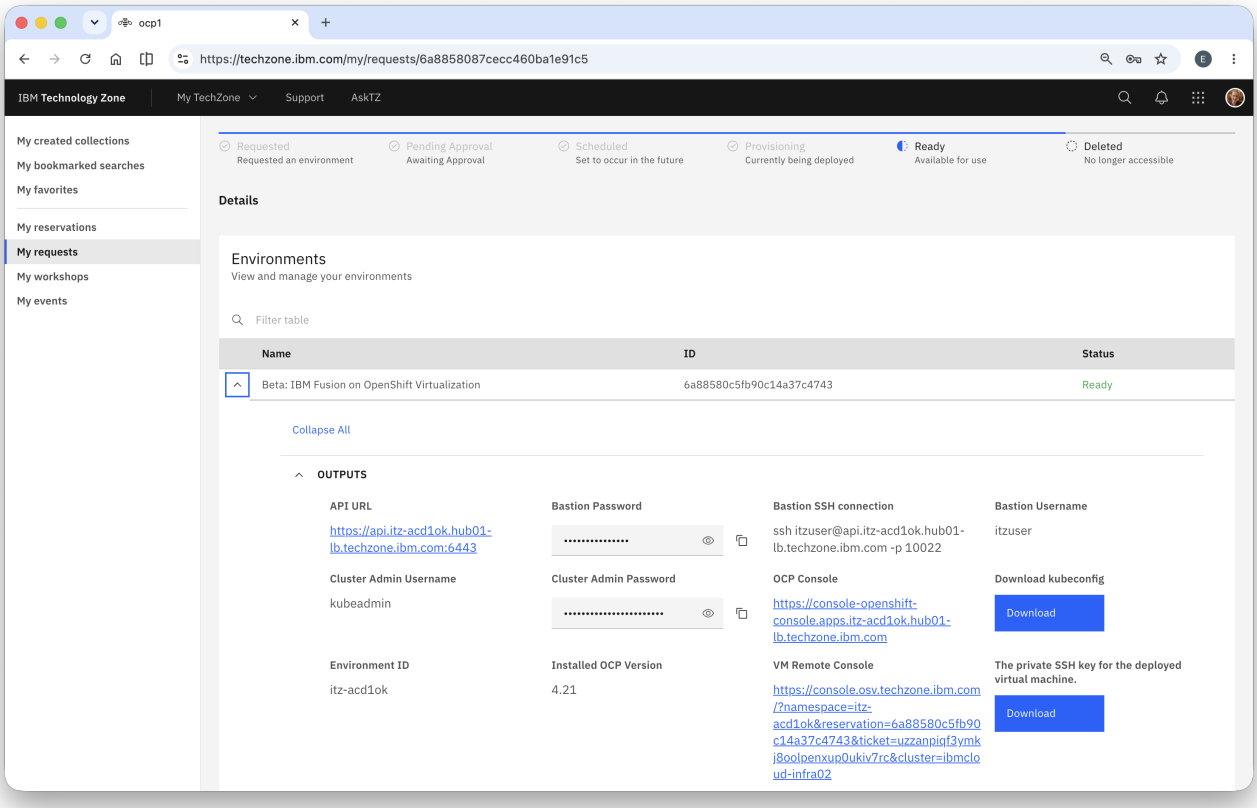
Once an environment reservation is successful you can start using it. If you are a first time then follow the steps below to get started.

Gather up some details

Start on the My requests page, locate the reservation you want to use, and make sure it is in the green "Reservation - Ready" state.

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Open the reservation page by clicking Open this reservation on the reservation tile (or the blue "eye" icon if you are in a list view instead of a tiled view.) Now expand the environment view by clicking the down-pointing arrow next to you environment name.



Before you continue we advise your to collects the various parameters of your environment such as passwords and access key file into a folder on your local desktop system. Create a folder which will be your work folder and collect the following artifacts into it:

- Download the kubeconfig file and save it
- Download the private SSH key and save it
- Create a text file and save the following parameters to it:
  - 24-character environment ID
  - Bastion user `itzuser` and the password from the hidden field
  - OCP cluster user `kubeadmin` and the password from the hidden field

For detailed instructions on connecting to the Web Console, SSH-ing to the Bastion, installing `oc` locally, and VM IP address references, see Accessing the Environment.

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Accessing Your TechZone Lab Environment

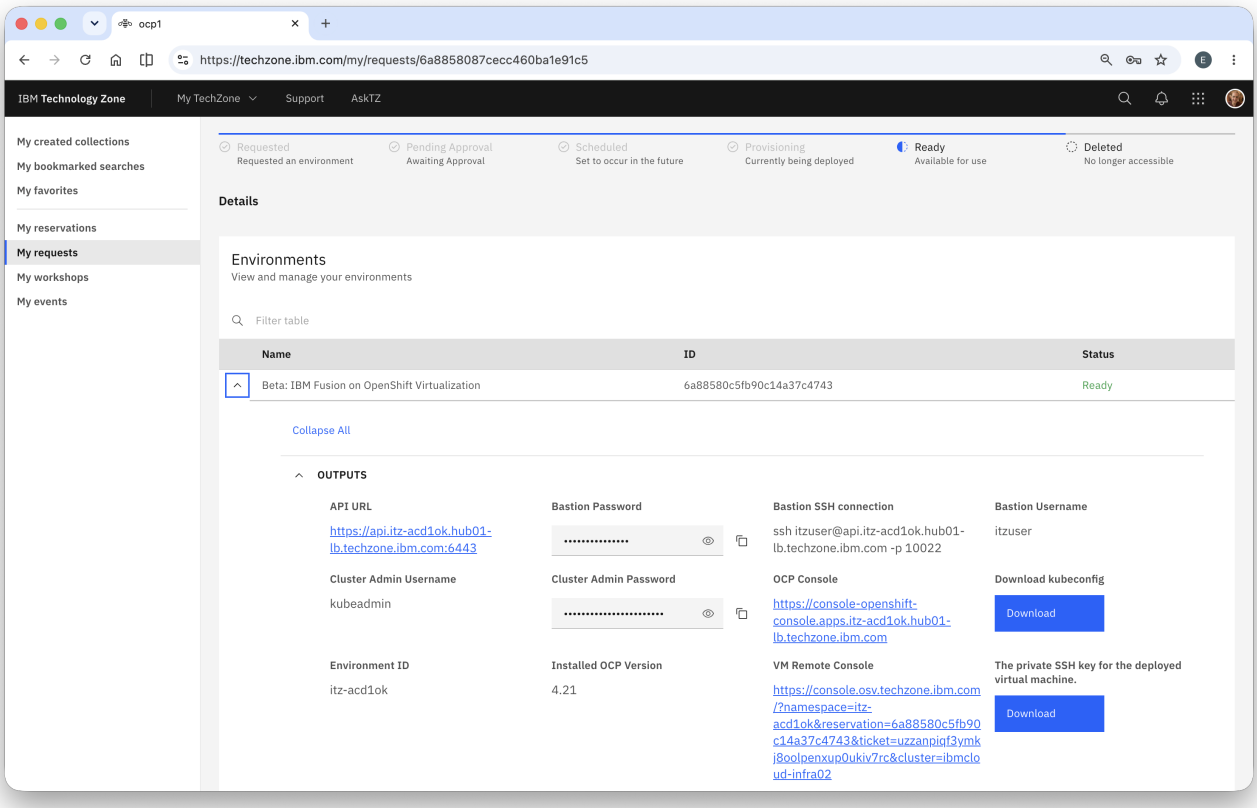
Introduction

Once your TechZone reservation is in the **Ready** state, you have three ways to interact with your OpenShift cluster. Most learners use a combination — the Web Console for UI-driven tasks and either the Bastion or a local `oc` install for CLI work.

Note If you have not yet provisioned a TechZone environment, complete the TechZone Quick Start first.

Gathering Your Credentials

All endpoints and credentials are found in the **Outputs** section of your reservation page. Navigate to My requests, open your reservation, and expand the environment row to reveal the Outputs panel shown below.



Before proceeding, collect the following into a working folder on your local machine:

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- Download the **kubeconfig** file — saves as `conf_kubeconfig_download.conf`
- Download the **private SSH key** — saves as `vm_ssh_key.vm` (check your Downloads folder)
- Note or copy the following values from the hidden fields (click the eye icon to reveal):
  - **Cluster Admin Password** (username is `kubeadmin`)
  - **Bastion Password** (username is `itzuser` — used for `sudo` on the Bastion, not SSH login)
- Note the **Bastion SSH connection** command — it is pre-built with the correct hostname and port
- Note the **API URL** — used for `oc login`

The following table describes every field in the Outputs panel:

Field	Description
API URL	OpenShift API endpoint used for `oc login` and kubeconfig
OCP Console	OpenShift Web Console URL — open directly in your browser
Cluster Admin Username	`kubeadmin`
Cluster Admin Password	Hidden — click the eye or copy icon
Bastion SSH connection	Pre-built SSH command — copy and paste directly into your terminal
Bastion Username	`itzuser`
Bastion Password	Hidden — required only for `sudo` on the Bastion, not for SSH login
Download kubeconfig	Download — saves as `conf_kubeconfig_download.conf`
The private SSH key	Download — required to authenticate SSH to the Bastion
VM Remote Console	Browser-based VNC access to the underlying cluster VMs
Environment ID	Your unique 24-character environment identifier
Installed OCP Version	The OpenShift version running in your cluster

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Important The Bastion host uses **key-based SSH authentication only** on port 10022. Password authentication is disabled on that port. You must download the private SSH key before attempting to connect.

Choosing an Access Method

Method	Best for	Requires
OpenShift Web Console	UI-driven tasks, monitoring, Software Catalog	Browser only
Bastion host (SSH)	CLI tasks — `oc` is pre-installed, no local setup needed	SSH client + private key
Local `oc` on your laptop	CLI tasks without connecting to the Bastion	`oc` installed + kubeconfig

Option 1: OpenShift Web Console

The OpenShift Web Console runs entirely in your browser — no software installation required.

1. From your TechZone reservation page, copy the **OCP Console** URL.
2. Open it in your browser.
3. On the login page, select the **kube:admin** identity provider.

   Tip Select **kube:admin** rather than IBMid to ensure you log in with full `cluster-admin` privileges.

4. Enter `kubeadmin` as the username and paste the **Cluster Admin Password** from your reservation outputs.
5. Upon successful authentication the OpenShift Administrator Dashboard is displayed.

Tip When your browser offers to save the password, accept — you will be logging back in frequently throughout the labs.

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Option 2: Bastion Host via SSH

The Bastion host has `oc`, `kubectl`, `helm`, and `curl` pre-installed. Connecting via SSH is the quickest way to run CLI commands without any local setup.

Step 1: Download the private SSH key

On your TechZone reservation page, click Download next to **The private SSH key for the deployed virtual machine**. The file saves as `vm_ssh_key.vm` (check your Downloads folder).

Step 2: Set key permissions

SSH requires the private key file to be readable only by your user.

macOS / Linux:

```
chmod 400 ~/Downloads/vm_ssh_key.vm
```

Windows (PowerShell):

```
icacls "$env:USERPROFILE\Downloads\vm_ssh_key.vm" /inheritance:r /grant:r
"$env:USERNAME:(R)"
```

Step 3: Connect via SSH

Copy the **Bastion SSH connection** command from your reservation page — it is pre-populated with the correct hostname and port. Add the `-i` flag pointing to your downloaded key:

macOS / Linux:

```
ssh -i ~/Downloads/vm_ssh_key.vm itzuser@api.<env-id>.techzone.ibm.com -p 10022
```

Windows (PowerShell — OpenSSH is built in to Windows 10 and 11):

```
ssh -i "$env:USERPROFILE\Downloads\vm_ssh_key.vm" itzuser@api.<env-id>.techzone.ibm.com -p
10022
```

Note If you are on an older Windows version without built-in OpenSSH, use PuTTY. Convert `vm_ssh_key.vm` to PuTTY format using **PuTTYgen** (Load → Save private key), then connect via PuTTY on port 10022.

Step 4: Log in to OpenShift from the Bastion

Once connected to the Bastion, log in to your OpenShift cluster using the **API URL** and **Cluster Admin**

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Password from your reservation:

```
oc login -u kubeadmin -p <cluster-admin-password> https://api.<env-
id>.techzone.ibm.com:6443
```

Verify connectivity:

```
oc whoami
oc get nodes
```

`oc whoami` should return `kubeadmin` and `oc get nodes` should list all cluster nodes in `Ready` state.

Option 3: Local `oc` CLI on Your Laptop

If you prefer to run `oc` commands directly from your laptop without SSH-ing to the Bastion, install the `oc` CLI and use the downloaded kubeconfig.

Step 1: Install the `oc` CLI

macOS:

```
# Using Homebrew
brew install openshift-cli

# Verify
oc version
```

Alternatively, download the binary from the OCP Web Console: click the Help icon (?) in the top-right masthead and select Help › Command Line Tools, then download the macOS binary and move it to `/usr/local/bin/`.

Linux:

```
# Download from the OCP Web Console (Help > Command Line Tools), or use the mirror:
curl -LO https://mirror.openshift.com/pub/openshift-v4/clients/ocp/stable/openshift-
client-linux.tar.gz
tar -xvf openshift-client-linux.tar.gz
sudo mv oc /usr/local/bin/
oc version
```

Windows:

Download the `oc` binary from the OCP Web Console (Help icon → Help › Command Line Tools → **Windows**). Extract the zip and move `oc.exe` to a folder on your `PATH` (e.g. `C:\tools\`).

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Alternatively, install via Chocolatey:

```
choco install openshift-cli
oc version
```

Step 2: Download the kubeconfig

On your TechZone reservation page, click Download next to **Download kubeconfig**. The file saves as `conf_kubeconfig_download.conf`.

Step 3: Connect using the kubeconfig

macOS / Linux:

```
export KUBECONFIG=~/Downloads/conf_kubeconfig_download.conf
oc whoami
oc get nodes
```

Windows (PowerShell):

```
$env:KUBECONFIG = "$env:USERPROFILE\Downloads\conf_kubeconfig_download.conf"
oc whoami
oc get nodes
```

A successful connection returns `system:admin` from `oc whoami`.

Tip

To make the kubeconfig permanent across terminal sessions, merge it into your default `~/.kube/config`:

```
# macOS / Linux
KUBECONFIG=~/.kube/config:~/Downloads/conf_kubeconfig_download.conf \
  oc config view --flatten > /tmp/merged && mv /tmp/merged ~/.kube/config
```

SSH to IBM Storage Scale node

Option 1: Bastion node

To SSH from the Bastion directly into the IBM Storage Scale node:

```
ssh itzuser@10.10.10.202 □
```

□ Verify this IP against the VM Remote Console page before use.

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Option 2: Directly from your workstation

macOS / Linux:

```
ssh -i ~/Downloads/vm_ssh_key.vm itzuser@api.<env-id>.techzone.ibm.com -p 10023
```

Windows (PowerShell — OpenSSH is built in to Windows 10 and 11):

```
ssh -i "$env:USERPROFILE\Downloads\vm_ssh_key.vm" itzuser@api.<env-id>.techzone.ibm.com -p 10023
```

Note If you are on an older Windows version without built-in OpenSSH, use PuTTY. Convert `vm_ssh_key.vm` to PuTTY format using **PuTTYgen** (Load → Save private key), then connect via PuTTY on port 10023.

VM IP Address Reference

The following IP addresses are assigned to VMs in the standard TechZone Fusion environment. Your configuration may include additional worker nodes — treat this as a representative reference:

VM	IP Address
`itz-*-bastion-1`	`10.10.10.201`
`itz-*-scale-*`	`10.10.10.202`
`itz-*-master-1`	`10.10.10.11`
`itz-*-master-2`	`10.10.10.12`
`itz-*-master-3`	`10.10.10.13`
`itz-*-infra-1`	`10.10.10.2`
`itz-*-infra-2`	`10.10.10.3`
`itz-*-infra-3`	`10.10.10.4`
`itz-*-worker-1`	`10.10.10.21`

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VM	IP Address
`itz-*-worker-2`	`10.10.10.22`
`itz-*-worker-3`	`10.10.10.23`

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IBM Fusion Operator Installation & Console Overview

Introduction

Although your Red Hat OpenShift cluster is provisioned on IBM Technology Zone, IBM Fusion has not yet been deployed. In this guide, you will install the IBM Fusion Operator from the Software Catalog, accept the product license, and explore the core administration interface.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Installing the IBM Fusion Operator

The IBM Operator Catalog and entitlement secrets are pre-configured on your TechZone OpenShift cluster.

Install via OpenShift Web Console

1. Log in to the OpenShift Web Console as `kubeadmin`.
2. In the left navigation menu, go to Ecosystem › Software Catalog.
3. In the search field, type `Fusion` and select the **IBM Storage Fusion** operator tile.
4. Click Install.
5. Leave all installation options as default:
   - **Update channel:** Stable channel (e.g., `v2.x`)
   - **Installation mode:** All namespaces on the cluster (default)
   - **Installed Namespace:** `ibm-spectrum-fusion-ns` (or Operator-recommended namespace)
   - **Approval Strategy:** Automatic
6. Click Install at the bottom of the form.
7. Wait until the Operator status shows **Succeeded** / **Installed operator - ready for use**.

Optional: Install via OpenShift CLI (`oc`)

You can also verify or initiate operator installation from the Bastion terminal:

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```
oc get csv -n ibm-spectrum-fusion-ns
```

Connecting to the IBM Fusion Management Console

IBM Fusion is container-native and integrates directly into the OpenShift console.

1. From the top masthead of the OpenShift Web Console, click the **Application Menu** icon (the 9-square grid).
2. Click **IBM Storage Fusion** from the dropdown menu. A new browser tab opens to the Fusion interface.
3. If prompted with the OpenShift login page, select **kube:admin** and enter your cluster credentials.
4. On first launch, review and select **I have read and accept the license agreement**, then click Continue.

> **Note** Accepting the license initializes the core `SpectrumFusion` Custom Resource (CR) in the `ibm-spectrum-fusion-ns` namespace.

5. You are redirected to the **Welcome to IBM Fusion** dashboard.

Navigating the IBM Fusion Dashboard

The IBM Fusion interface provides a unified view of data management services:

- **Services:** View installed services and install optional capabilities (Fusion Data Foundation, Backup & Restore, Data Cataloging).
- **Storage:** Manage local storage clusters, StorageClasses, and remote IBM Storage Scale file systems.
- **Backup & Restore:** Configure backup locations (S3), retention policies, and application protection.
- **Events:** View Kubernetes-native events with severity filtering (Info, Warning, Critical).
- **Settings:** Review cluster networking, Call Home diagnostics, and cluster interconnect status.
- **Help & Support:** Access the question-mark icon in the masthead to download support logs or link to official IBM documentation.

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Configuring Fusion Data Foundation (FDF)

Introduction

In this guide you will enable the Data Foundation service in the IBM Fusion Console and configure the local Ceph storage cluster using the OpenShift Web Console wizard.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Enabling the Data Foundation Service

Note This guide covers the **Local** configuration mode, the standard mode for single-cluster labs. For an overview of all FDF configuration modes (Local, Dynamic, External, and Provider), see Persistent Storage: Architecture & Concepts.

1. In the IBM Fusion Console, navigate to **Services** in the left navigation menu.
2. Locate the **Data Foundation** service tile, click the tile, and then click Install.
3. On the **Install service** screen, select the **Local** tile and click Install.

   Note Selecting **Local** automatically installs the Local Storage Operator (LSO) — no separate LSO installation step is required.

4. Wait for the service status to show **Healthy**.

Configuring the Ceph Storage Cluster

Once the Data Foundation service is enabled, configure the local Ceph storage cluster through the OpenShift Web Console wizard launched from Fusion:

Step 1: Launch Storage Cluster Creation

1. In the IBM Fusion Console, navigate to Storage › Data Foundation (or **Local storage**).
2. Click Get Started (or Configure Data Foundation / Create Storage Cluster).

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3. This redirects to the **Create StorageSystem** wizard in the OpenShift Web Console.

Step 2: Backing Storage

1. On the **Backing storage** page, select **Create a new StorageClass using local storage devices**.
2. Click Next.

Step 3: Advanced Settings

1. On the **Advanced Settings** page, configure the default StorageClass options:
   - Leave **Use Ceph RBD as the default StorageClass** checked. This sets Ceph RBD as the default StorageClass, eliminating the need to manually annotate a StorageClass after installation.
   - If this cluster will host virtual machines, also check **Set default StorageClass for virtualization**. This marks the RBD virtualization StorageClass as the default for KubeVirt VM disks after installation.
2. Leave all other options at their defaults and click Next.

Note Both default StorageClass options are configured here in the wizard. No additional manual annotation steps are required after installation.

Step 4: Create Local Volume Set

1. Enter `ibm-spectrum-fusion` as the **LocalVolumeSet name**.
2. Select **Disks on selected nodes** and choose the dedicated infra nodes (`itz-*-infra-1`, `itz-*-infra-2`, `itz-*-infra-3`).
3. Set **Disk Type** to **SSD/NVMe**.
4. Click the `^` to reveal the **Advanced** section, and enter "100" in the minimum drive size field; this will exclude the 50GiB drives.
5. Click Next, then click Yes to confirm creation of the LocalVolumeSet.

Step 5: Capacity and Nodes

1. Verify the **Available raw capacity** populated from the discovered disks.
2. Select a **resource profile** — **Balanced** is recommended for lab environments.
3. Click Next.

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Step 6: Security and Network

1. Keep the default encryption settings unless KMS integration (Vault or Thales CipherTrust) is required.
2. Keep the default network option (**Default (OVN)**) unless using Multus.
3. Click Next.

Note If you are configuraing storage for Hosted Control Plane (HCP) then select the "Host" radio button.

Step 7: Data Protection

1. If this cluster will be configured for Regional Disaster Recovery, select **Prepare cluster for disaster recovery (Regional-DR only)**.
2. Otherwise leave unchecked and click Next.

Step 8: Review and Create

1. Review the configuration summary. Click Back to correct any settings.
2. Click Create StorageSystem.

Note Storage cluster initialization takes approximately 10–15 minutes. Wait until the cluster status reports **Ready / Healthy** under Storage › Data Foundation in the OpenShift console before proceeding.

Step 9: Verify StorageClasses and Ceph Cluster Health

From the OpenShift Web Console:

1. Navigate to Storage › StorageClasses and confirm the standard FDF storage classes are listed.
2. Navigate to Storage › Data Foundation and confirm the storage cluster status shows **Ready / Healthy**.
3. Navigate to Ecosystem › Installed Operators › OpenShift Data Foundation and confirm the operator status shows **Succeeded**.

From the CLI:

```
# Check provisioned StorageClasses
oc get sc

# Verify Ceph cluster status
```


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```
oc get storagecluster -n openshift-storage

# Confirm all ODF pods are Running or Completed
oc get pods -n openshift-storage
```

The following standard StorageClasses should now be available:

- `ocs-storagecluster-ceph-rbd` (Block storage for containers & VMs)
- `ocs-storagecluster-cephfs` (Shared file storage / RWX)
- `openshift-storage.noobaa.io` (Object Bucket Claims / S3)

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IBM Fusion Regional Disaster Recovery (RDR): Architecture & Concepts

Introduction

Regional Disaster Recovery (RDR) provides asynchronous data replication and cross-cluster workload failover between two geographically separated OpenShift clusters. It is composed of Red Hat Advanced Cluster Management for Kubernetes (RHACM) and IBM Fusion Data Foundation components to provide application and data mobility across OpenShift Container Platform clusters.

While **Metro Disaster Recovery (MDR)** operates over low-latency (<10ms) synchronous networks using stretch clusters, **Regional DR** is designed for independent clusters across metropolitan or continental distances where network latency precludes synchronous writes. Because replication is asynchronous, there is a potential for a small amount of data loss in an unplanned outage, but protection covers a broad set of failure scenarios.

This solution requires Red Hat Advanced Cluster Management (RHACM) and Fusion Data Foundation Advanced entitlements.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Core Technology Components

Red Hat Advanced Cluster Management (RHACM)

RHACM provides the ability to manage multiple clusters and application lifecycles, serving as the control plane in a multi-cluster environment. It is split into two parts:

- **RHACM Hub** — Runs on the multi-cluster control plane and provides centralized orchestration, policy governance, and application lifecycle management across all managed clusters.
- **Managed Clusters** — Run the RHACM agent components that receive and execute tasks from the Hub.

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Red Hat OpenShift GitOps

Red Hat OpenShift GitOps integrates declarative, Git-driven configuration and application management powered by Argo CD. In an RDR environment, OpenShift GitOps is a prerequisite for the Multicluster Orchestrator to support multi-cluster coordination and application discovery across managed clusters.

Fusion Data Foundation (FDF)

FDF provides the storage and replication layer for RDR. It is backed by Ceph (managed by Rook) and enhanced with the following disaster recovery capabilities:

- Enable RBD block pools for mirroring across FDF instances on different clusters.
- Mirror specific images within an RBD block pool asynchronously.
- Provide CSI add-ons to manage per-PVC mirroring, giving fine-grained control over which persistent volumes are replicated.

OpenShift DR (ODR) Operators

OpenShift DR orchestrates disaster recovery for stateful applications across peer OpenShift clusters. It provides cloud-native interfaces to manage the lifecycle of application state on Persistent Volumes. ODR is split into three components:

- **FDF Multicluster Orchestrator** — Installed on the Hub cluster alongside RHACM. Orchestrates configuration and peering of FDF clusters for Metro and Regional DR relationships.
- **FDF DR Hub Operator** — Automatically installed as part of the Multicluster Orchestrator on the Hub cluster. Orchestrates failover and relocation of DR-enabled applications.
- **FDF DR Cluster Operator** — Automatically installed on each managed cluster in a DR relationship. Manages the lifecycle of all PVCs belonging to a protected application.

Submariner

Submariner establishes direct, encrypted cross-cluster pod-to-pod and service-to-service networking between the primary and secondary clusters, required for ODR and RHACM to communicate across cluster boundaries.

Managed vs. Discovered Applications

RHACM supports two application models for disaster recovery:

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- **Managed applications** are deployed and managed through the ACM application model using a multi-cluster application definition (GitOps / Argo CD). ACM tracks and deploys these applications declaratively across clusters. OpenShift API for Data Protection (OADP) is *not* required for managed applications because the application definition lives in source control and is re-applied during failover.
- **Discovered applications** already exist on managed clusters outside of the ACM application model. ACM discovers these through OpenShift GitOps or Argo CD operators and provides visibility, health monitoring, and DR integration. OADP is deployed alongside ACM to back up and restore Kubernetes objects during failover or relocation.

Note The RDR lab uses the **Discovered applications** model. OADP is installed on both clusters to enable Kubernetes object backup and restore in conjunction with FDF volume replication.

Disaster Recovery Operations: Failover vs. Relocate

Failover (Unplanned Outage)

Failover switches an application and its state from the primary cluster to the secondary cluster when the primary becomes unavailable. The process is administrator-initiated — the DR solution orchestrates application restart on the secondary cluster using the most recent asynchronously replicated storage snapshot. Because replication is asynchronous, a small amount of data loss is possible depending on replication lag at the time of the outage.

Relocate (Planned Migration / Failback)

Relocate moves an application back to the original primary cluster after the disaster is resolved and the primary is restored. Because the process is planned, data synchronization completes before the application is restarted on the primary site, ensuring zero data loss.

	Failover	Relocate
Trigger	Unplanned primary outage	Planned — primary restored
Direction	Primary → Secondary	Secondary → Primary

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	Failover	Relocate
Data loss	Possible (async replication lag)	None (synchronized before cutover)
Initiation	Administrator-initiated	Administrator-initiated

Lab Architecture

The RDR lab uses two independent OpenShift clusters:

- **`local-cluster` (Primary)** — Hosts the primary workloads, Fusion Data Foundation, and Red Hat Advanced Cluster Management.
- **`ocp2` (Secondary / DR site)** — Configured as the disaster recovery site with its own FDF instance.

Important

In this lab, ACM is deployed on `local-cluster` rather than on a separate hub cluster. This configuration is not supported for production Regional Disaster Recovery and is intended solely for learning purposes. In a production environment, RHACM should be deployed on a dedicated hub cluster that does not participate in data replication.

Software Versions

The following software versions were used to develop and validate the RDR lab:

Component	Version
Red Hat OpenShift Container Platform (Primary)	4.21.33
Red Hat OpenShift Container Platform (Secondary)	4.21.33
Red Hat Advanced Cluster Management	2.17.3
Red Hat OpenShift Virtualization	4.21.17
IBM Fusion	2.14

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Component	Version
OpenShift Application Data Protection (OADP)	1.5.8

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Regional Disaster Recovery (RDR): Environment Preparation & Setup

Introduction

Implementing **Regional Disaster Recovery (RDR)** with IBM Fusion and Red Hat Advanced Cluster Management (ACM) requires two distinct OpenShift clusters deployed across separate network subnets, connected via Submariner, with all default TechZone ACM management disconnected.

Reserving Two TechZone Environments (OCP-V)

Regional DR requires two independent clusters provisioned on IBM Technology Zone (TechZone) using the **IBM Fusion on OpenShift Virtualization** environment.

Follow the reservation workflow outlined in the TechZone Quick Start Guide, paying close attention to the **Configuration** step where network settings must be customized.

Cluster 1: Primary (`local-cluster`)

When reserving the primary cluster:

1. **Environment Tile:** Select **IBM Fusion on OpenShift Virtualization**.
2. **Name:** `local-cluster`.
3. **Purpose & Deployment:** Set your intended business purpose and preferred geography.
4. **Configuration (Network Defaults):** Keep the primary cluster on the default network assignments:
    - **VM Network CIDR:** `10.10.10.0/24` (default)
    - **OCP/Kubernetes Cluster Network:** `10.128.0.0/14` (default)
    - **OCP/Kubernetes Service Network:** `172.30.0.0/16` (default)

Cluster 2: Secondary / DR Site (`ocp2`)

Important **Non-Overlapping Subnets Required for Submariner:** Submariner cross-cluster networking requires that the VM Network CIDR, OCP/Kubernetes Cluster Network, and OCP/Kubernetes Service Network do not overlap between the two clusters.

When reserving the secondary cluster, click Customize on the **Configuration** step and modify the network

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dropdowns:

- **Name:** `ocp2`
- **VM Network CIDR:** Select an alternate subnet by incrementing the 3rd octet (`10.10.11.0/24`, `10.10.12.0/24`, or `10.10.13.0/24`).
- **OCP/Kubernetes Cluster Network:** Select one of the alternate options: `10.140.0.0/14`, `10.144.0.0/14`, or `10.152.0.0/14`.
- **OCP/Kubernetes Service Network:** Select one of the alternate options: `172.33.0.0/16`, `172.34.0.0/16`, or `172.35.0.0/16`.

Network Subnet Comparison Matrix (OCP-V)

Network Parameter	Primary Cluster (`local-cluster`) [Defaults]	Secondary Cluster (`ocp2`) Selection Options
VM Network CIDR	`10.10.10.0/24`	`10.10.11.0/24`, `10.10.12.0/24`, or `10.10.13.0/24`
OCP/Kubernetes Cluster Network	`10.128.0.0/14`	`10.140.0.0/14`, `10.144.0.0/14`, or `10.152.0.0/14`
OCP/Kubernetes Service Network	`172.30.0.0/16`	`172.33.0.0/16`, `172.34.0.0/16`, or `172.35.0.0/16`

Disconnecting TechZone ACM

TechZone automatically attaches reserved OpenShift clusters to a central TechZone ACM management hub. Because you will be deploying your own ACM instance on `local-cluster`, you must disable TechZone ACM auto-import and clean up the existing klusterlet agents.

Important These steps need to be completed prior to adding any additional services as the cleanup script will remove some artifacts needed by them.

Step 1: Disable TechZone ACM Auto-Import

1. In the TechZone reservation page, open the **VM Remote Console** for both reservations (`local-cluster` and `ocp2`).

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2. Under **Cluster management**, click Techzone ACM.
3. Toggle the **ACM Auto-Import** switch to **Disabled**.
4. Allow 15–30 minutes for TechZone to reconcile and detach the cluster.

Step 2: Remove TechZone ACM Klusterlet

On **both clusters** (`local-cluster` and `ocp2`), log in to the Bastion host as `root` (`sudo -i`) and execute the cleanup script:

Run cleanup script to remove TechZone ACM artifacts

```
bash <(curl -s https://itz-ocpv-downloads.s3.us-east.cloud-object-
storage.appdomain.cloud/scripts/cleanup-managed-cluster.sh)
```

Verify that the klusterlet CR is removed

```
oc get klusterlet
```

Ensure the command returns `No resources found`.

RDR Lab Prerequisites

The following steps must be completed on **both** clusters (`local-cluster` and `ocp2`) before configuring RHACM. Each step is identical on both clusters — you are running the same Fusion install and FDF setup on two separate TechZone reservations. To save time, run them in parallel on each cluster.

Install the IBM Fusion Operator

IBM Fusion has not yet been installed on your provisioned TechZone OpenShift environments. Follow the steps in IBM Fusion Operator Installation & Console Overview to install the operator and accept the license agreement on both clusters.

Install Fusion Data Foundation

Install and configure FDF in **Local** mode on both clusters, following Configuring Fusion Data Foundation (FDF).

Install OpenShift Application Data Protection (OADP)

OADP is required on both clusters to back up and restore Kubernetes objects alongside FDF volume replication during failover and relocation.

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1. In the OpenShift Web Console, navigate to Ecosystem › Software Catalog.
2. Search for `OADP` and select the **OADP Operator** tile.
3. Click Install, keep all default settings, and click Install.
4. Wait for the operator status to show **Succeeded** under Ecosystem › Installed Operators.

Verify from the CLI:

```
oc get csv -n openshift-adp | grep oadp
```

The `PHASE` column should show `Succeeded` on both clusters before continuing.

5. Click the Quick create icon (the `+` icon in the masthead) and select **Import YAML**.
6. Copy the following YAML, paste it into the editor, and click Create:

```
apiVersion: oadp.openshift.io/v1alpha1
kind: DataProtectionApplication
metadata:
  labels:
    app.kubernetes.io/component: velero
  name: velero
  namespace: openshift-adp
spec:
  backupImages: false
  configuration:
    nodeAgent:
      enable: false
      uploaderType: restic
    velero:
      defaultPlugins:
        - openshift
        - aws
      noDefaultBackupLocation: true
```

7. A `DataProtectionApplication` resource is created in the `openshift-adp` project on each cluster.

Perform all prerequisite steps on **both clusters** before continuing.

Preparing Submariner Cross-Cluster Gateways

Submariner establishes secure cross-cluster pod-to-pod networking and requires one node in each cluster designated as the gateway.

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Important The TechZone environment forwards Submariner traffic exclusively through `itz-*-worker-1`. You must label `itz-*-worker-1` on both clusters — using any other worker node will cause Submariner to fail to establish a connection.

Step 1: Label Gateway Nodes on Both Clusters

Run the following command on **both `local-cluster` and `ocp2`**:

Label `itz-*-worker-1` as the Submariner gateway node

```
oc get node | grep "worker-1 " | cut -d " " -f1 | xargs -I {} oc label node {}
submariner.io/gateway=true
```

Step 2: Annotate Primary Gateway (Run on `local-cluster` only)

Retrieve your hub API hostname, resolve its public IP, and annotate `itz-*-worker-1`:

Resolve API server public IP

```
oc whoami --show-server | cut -d "/" -f3 | cut -d ":" -f1 | xargs -I {} nslookup {}
1.1.1.1
```

Annotate `itz-*-worker-1` with preferred server and public IP. Replace `<resolved-public-ip>` below with Address discovered above.

```
oc get node | grep "worker-1 " | cut -d " " -f1 | xargs -I {} oc annotate node {} \
  gateway.submariner.io/preferred-server=true \
  gateway.submariner.io/public-ip=ipv4:<resolved-public-ip>
```

Configuring RHACM

The following steps are performed on `local-cluster` only.

Install Red Hat Advanced Cluster Management

1. In the OpenShift Web Console of `local-cluster`, navigate to Ecosystem › Software Catalog.
2. Search for `Advanced Cluster Management for Kubernetes` and select the operator tile.
3. Click Install, keep all default settings, and click Install.
4. Wait for the Create MulticlusterHub button to become active, then click it.

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5. Keep all default settings and click Create.
6. The `multiclusterhub` status will progress through `Phase: Installing` to `Phase: Running`.
7. After the console reloads, the **Fleet Management** interface is available via Core Platform › Fleet Management.

Install Red Hat OpenShift GitOps

Red Hat OpenShift GitOps is required prior to installing the Multicluster Orchestrator.

1. In the OpenShift Web Console of `local-cluster`, navigate to Ecosystem › Software Catalog.
2. Search for `Red Hat OpenShift GitOps` and select the operator tile.
3. Click Install, accept all default settings, and click Install.
4. Wait for the operator status to display **Succeeded** under Ecosystem › Installed Operators.

Install Multicluster Orchestrator

The Multicluster Orchestrator provides the FDF DR Hub Operator and enables the Disaster Recovery policy interface in RHACM.

1. Navigate to Ecosystem › Software Catalog.
2. Search for `Multicluster Orchestrator` and select the operator tile `Provided by IBM`.
3. Click Install.
4. Under the **Console plugin** section, click Enable, then scroll down and click Install.
5. When the **Web console update is available** popup appears, click **Refresh web console**.

Import the Secondary Cluster (`ocp2`)

1. Download the kubeconfig for `ocp2` from its TechZone reservation page (Download kubeconfig). Rename the file to `ocp2_kubeconfig.conf`.
2. Open the file in a text editor and copy all of its contents.
3. In the `local-cluster` OpenShift Web Console, navigate to Core Platform › Fleet Management and click **All Clusters**.
4. Click Import cluster (or Import an existing cluster from the getting started popup).
5. Enter `ocp2` in the **Name** field and set **Cluster set** to `default`.
6. From the **Import mode** dropdown, select **Kubeconfig**.

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7. Paste the contents of `ocp2_kubeconfig.conf` into the **Kubeconfig** field and click Next.
8. Click Next through any remaining steps, then click Import.
9. Wait for the cluster status to show **Ready** with a green checkmark.

Install Submariner Add-ons

1. In Fleet Management, navigate to Infrastructure › Clusters and select the **Cluster sets** tab.
2. Click the **default** cluster set to open its overview.
3. Select the **Submariner add-ons** tab and click Install Submariner add-ons.
4. In the **Target clusters** selection box, select both `local-cluster` and `ocp2`, then click Next.
5. Click Install.
6. When running OVN-Kubernetes in local gateway mode, apply the health-check workaround on **both clusters** after installing the Submariner add-on:

   ```
   oc apply -f https://itz-ocpv-downloads.s3.us-east.cloud-object-
   storage.appdomain.cloud/scripts/submariner-ovn-snat-workaround.yaml
   ```

   Verify DaemonSet status:

   ```
   oc get ds -n submariner-operator submariner-ovn-snat-fix
   ```
7. In the **Submariner add-ons** table, verify that both `local-cluster` and `ocp2` display:
   - **Connection status:** `Healthy`
   - **Agent status:** `Healthy`
   - **Gateway nodes labeled:** `Nodes labeled`

Note The connection status may briefly show **Degraded** while cross-cluster tunnels are being established. Allow a few minutes for the status to turn **Healthy**.

Create a Disaster Recovery Policy

1. In Fleet Management, navigate to Data Services › Disaster Recovery.
2. Click Create a disaster recovery policy.
3. Enter `ocp1-ocp2-dr` in the **Policy name** field.
4. In **Connect clusters**, check both `local-cluster` and `ocp2`.

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5. **(Optional)** If Virtual Machines will be tested with RDR, click **Advanced Settings** and select **Enable disaster recovery support for restored and cloned PersistentVolumeClaims (For Data Foundation only)**.
6. Click Create.
7. Wait for the policy status to update from **Not validated** to **Validated**.

Verify the DR policy is validated from the CLI (run on `local-cluster`):

```
oc get drpolicy ocp1-ocp2-dr
```

The `VALIDATED` column should show `True` before proceeding to the failover lab.

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Simulating Regional Failover and Failback

Introduction

In this exercise, you will deploy a sample stateful application on `local-cluster`, protect it with a disaster recovery policy, and execute simulated failover and failback operations between the primary and secondary clusters.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Deploying a Sample Application

1. In the `local-cluster` OpenShift Web Console, navigate to Home › Projects and click Create Project.
2. Enter `filebrowser` as the project name and click Create.
3. Ensure the active project is set to `filebrowser`. Click the Quick create icon (the `+` icon in the masthead) and select **Import YAML**.
4. Copy the following YAML, paste it into the editor, and click Create:

```
kind: PersistentVolumeClaim
apiVersion: v1
metadata:
  labels:
    app: filebrowser
    app.kubernetes.io/component: filebrowser
    app.kubernetes.io/instance: filebrowser
    app.kubernetes.io/name: filebrowser
    app.kubernetes.io/part-of: filebrowser-app
  name: files
spec:
  storageClassName: ocs-storagecluster-ceph-rbd
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 10Gi
```

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```
---
kind: PersistentVolumeClaim
apiVersion: v1
metadata:
  labels:
    app: filebrowser
    app.kubernetes.io/component: filebrowser
    app.kubernetes.io/instance: filebrowser
    app.kubernetes.io/name: filebrowser
    app.kubernetes.io/part-of: filebrowser-app
  name: filebrowser-config
spec:
  storageClassName: ocs-storagecluster-ceph-rbd
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 1Gi

---
kind: ConfigMap
apiVersion: v1
metadata:
  name: my-filebrowser-config
  labels:
    app: filebrowser
    app.kubernetes.io/component: filebrowser
    app.kubernetes.io/instance: filebrowser
    app.kubernetes.io/name: filebrowser
    app.kubernetes.io/part-of: filebrowser-app
data:
  .filebrowser.json: |
    {
      "port": 8080,
      "baseURL": "",
      "address": "",
      "log": "stdout",
      "database": "/config/database.db",
      "root": "/files"
    }
```


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```
---
kind: Deployment
apiVersion: apps/v1
metadata:
  name: filebrowser
  labels:
    app: filebrowser
    deployment: filebrowser
spec:
  replicas: 1
  selector:
    matchLabels:
      app: filebrowser
  template:
    metadata:
      labels:
        app: filebrowser
        app.kubernetes.io/component: filebrowser
        app.kubernetes.io/instance: filebrowser
        app.kubernetes.io/name: filebrowser
        app.kubernetes.io/part-of: filebrowser-app
    spec:
      restartPolicy: Always
      serviceAccountName: default
      schedulerName: default-scheduler
      enableServiceLinks: true
      terminationGracePeriodSeconds: 30
      securityContext: {}
      containers:
        - resources:
            requests:
              cpu: "100m"
              memory: "128Mi"
            limits:
              cpu: "500m"
              memory: "512Mi"
          readinessProbe:
            httpGet:
              path: /health
              port: 8080
            initialDelaySeconds: 2
            timeoutSeconds: 2
```

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```
      periodSeconds: 5
      successThreshold: 1
      failureThreshold: 3
    terminationMessagePath: /dev/termination-log
    name: filebrowser
    livenessProbe:
      httpGet:
        path: /health
        port: 8080
      initialDelaySeconds: 10
      timeoutSeconds: 2
      periodSeconds: 10
      successThreshold: 1
      failureThreshold: 3
    startupProbe:
      httpGet:
        path: /health
        port: 8080
      failureThreshold: 30
      periodSeconds: 10
    env:
      - name: TZ
        value: UTC
    ports:
      - name: http
        containerPort: 8080
        protocol: TCP
    imagePullPolicy: IfNotPresent
    securityContext:
      runAsNonRoot: true
      allowPrivilegeEscalation: false
      capabilities:
        drop:
          - ALL
    volumeMounts:
      - name: config
        mountPath: /config
      - name: filebrowser-config
        mountPath: /.filebrowser.json
        subPath: .filebrowser.json
      - name: data
        mountPath: /files
```


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```
          image: 'filebrowser/filebrowser:v2.18.0'
      automountServiceAccountToken: true
      serviceAccount: default
      volumes:
        - name: config
          persistentVolumeClaim:
            claimName: filebrowser-config
        - name: filebrowser-config
          configMap:
            name: my-filebrowser-config
            defaultMode: 420
        - name: data
          persistentVolumeClaim:
            claimName: files
      dnsPolicy: ClusterFirst
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxUnavailable: 1
      maxSurge: 1
  revisionHistoryLimit: 3
  progressDeadlineSeconds: 600

---
kind: Service
apiVersion: v1
metadata:
  name: filebrowser
  labels:
    app: filebrowser
    app.kubernetes.io/component: filebrowser
    app.kubernetes.io/instance: filebrowser
    app.kubernetes.io/name: filebrowser
    app.kubernetes.io/part-of: filebrowser-app
spec:
  ports:
    - name: 8080-tcp
      protocol: TCP
      port: 8080
      targetPort: 8080
  type: ClusterIP
  selector:
```

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```
    app: filebrowser

---
kind: Route
apiVersion: route.openshift.io/v1
metadata:
  name: filebrowser
  labels:
    app: filebrowser
    app.kubernetes.io/component: filebrowser
    app.kubernetes.io/instance: filebrowser
    app.kubernetes.io/name: filebrowser
    app.kubernetes.io/part-of: filebrowser-app
  annotations:
    openshift.io/host.generated: 'true'
spec:
  to:
    kind: Service
    name: filebrowser
    weight: 100
  port:
    targetPort: 8080-tcp
  tls:
    termination: edge
    insecureEdgeTerminationPolicy: Redirect
  wildcardPolicy: None
```

5. Verify the application pods and PVCs are running:

```
oc get pods,pvc -n filebrowser
```

Assigning a DR Policy to the Application

1. In the `local-cluster` Web Console, navigate to Core Platform › Fleet Management.
2. Go to Data Services › Disaster Recovery.
3. Click Enroll applications and select **ACM Discovered Applications**.
4. From the **DR cluster** dropdown, select `local-cluster`. Locate the `filebrowser` application in the `filebrowser` namespace, enter `filebrowser` in the **Name** field, and click Next.
5. Click Add label expression and configure the application label:

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- **Label / Key:** `app`
- **Values:** `filebrowser`

6. Click Add PVC label selector and configure the PVC label:
   - **Label / Key:** `app`
   - **Values:** `filebrowser`
7. Click Next.
8. Select the `ocp1-ocp2-dr` policy and click Next.
9. Review the summary and click Apply.

Verify the DR policy was applied from the CLI (run on `local-cluster`):

```
oc get drpc -n openshift-dr-ops
```

The `CURRENTSTATE` should show `Deployed` and `PREFERREDCLUSTER` should show `local-cluster` before proceeding.

Verifying Replication

Once the DR policy is assigned, FDF begins asynchronous replication of the application's PVCs to `ocp2`.

In the `local-cluster` Web Console (Core Platform › Fleet Management → Data Services › Disaster Recovery → **Protected applications**), verify that the DR status displays **Healthy**. Click the status badge to open the drawer and confirm that both **Application volumes (PVCs)** and **Kubernetes resources** display recent sync timestamps.

Preparing the Secondary Cluster for Failover

This section prepares the secondary cluster for failover or relocation by creating the project on `ocp2` with matching security settings from `local-cluster` to ensure seamless application transition without SELinux or UID permission issues.

1. In the `local-cluster` Web Console, navigate to Home › Projects and click the `filebrowser` project.
2. Click the **YAML** tab.
3. Select all the text in the YAML editor and copy it to the clipboard.
4. Switch to the `ocp2` Web Console, click the Quick create icon (the `+` icon in the masthead), and select **Import YAML**.

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5. Paste the copied YAML into the editor.
6. Remove the cluster-specific runtime metadata fields from the pasted YAML:
   - Under `metadata`: delete `uid`, `resourceVersion`, `creationTimestamp`, and `managedFields`.
   - At the bottom: delete `spec`, `finalizers`, and `status`.

   Note Ensure you preserve `apiVersion`, `kind`, `metadata.name`, and all `metadata.annotations` (including `openshift.io/sa.scc.mcs`, `openshift.io/sa.scc.supplemental-groups`, and `openshift.io/sa.scc.uid-range`) as well as `metadata.labels`.
7. Click Create to recreate the project on `ocp2` with identical security context and SELinux annotations.
8. On `local-cluster`, confirm the DRPolicy status is `Validated`:

```
oc get drpolicy ocp1-ocp2-dr
```

Executing a Failover (Simulated Unplanned Outage)

1. In the `local-cluster` Web Console, navigate to Core Platform › Fleet Management → Data Services › Disaster Recovery.
2. Click the **Protected applications** tab.
3. Locate the `filebrowser` application in the list.
4. Click the application actions menu (three stacked dots) and select **Failover application**.
5. In the **Failover application** modal dialog:
   - Verify **Target cluster** indicates `ocp2`.
   - Verify **Failover readiness** displays **Ready**.
6. Click Initiate.
7. In the **Disaster Recovery** table, click the status link (such as **FailingOver** or **WaitForUserCleanup**) in the **Data protection** column to view the failover activity drawer.

   Observe the 4 lifecycle progress steps: * **Preparing** — Prepares the target environment. * **Failover** — Directs storage replication cutover to `ocp2`. * **Restoring** — Restores Kubernetes resources and attaches the replicated PVCs. * **Clean up** / **WaitForUserCleanup** — Confirms application cutover on `ocp2`.

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8. Switch to the `ocp2` Web Console (or terminal) and observe the application pods and PVCs running in the `filebrowser` namespace:

```
oc get pods,pvc -n filebrowser -w
```

Preparing for Failback (User Cleanup on Failed Cluster)

Once failover completes, the DR activity drawer indicates: **"Clean up application resources on the failed cluster local-cluster to start the replication."** (status: `WaitForUserCleanup`). Lingering application resources must be removed from `local-cluster` to enable reverse replication from `ocp2` back to `local-cluster`.

1. Open a terminal connected to `local-cluster` and delete all application resources, ConfigMaps, and PVCs associated with the `filebrowser` application:

```
export NS=filebrowser
oc -n $NS delete all -l app=filebrowser
oc -n $NS delete cm -l app=filebrowser
oc -n $NS delete pvc -l app=filebrowser
```

2. In the `local-cluster` Web Console (Core Platform › Fleet Management → Data Services › Disaster Recovery → **Protected applications**), verify that the DR status returns to **Healthy**. Click the status badge and confirm:
   - **Message: All volumes & Kubernetes resources are synced**
   - **Application volumes (PVCs):** Displays a recent sync timestamp.
   - **Kubernetes resources:** Displays a recent sync timestamp.
   - **DR Policy:** `ocp1-ocp2-dr, sync every 5m`
3. Confirm DR status from the CLI on `local-cluster`:

```
oc get drpc -n openshift-dr-ops
```

   The output should confirm `DESIREDSTATE: Failover` and `CURRENTSTATE: FailedOver` targeting `ocp2`:

```
NAME          AGE   PREFERREDCLUSTER   FAILOVERCLUSTER   DESIREDSTATE   CURRENTSTATE
filebrowser   80m   local-cluster      ocp2              Failover       FailedOver
```

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Executing Failback (Relocate to Primary)

Unlike an unplanned failover, **Relocate** is a planned, zero-data-loss operation that transitions the workload gracefully back to `local-cluster`. Because `filebrowser` is a **Discovered application**, the running workload on `ocp2` must be deleted during the relocation process so its storage volume can be unmounted and final delta synchronization can complete.

1. In Fleet Management on `local-cluster`, navigate to Data Services › Disaster Recovery.
2. Click the **Protected applications** tab and locate the `filebrowser` application.
3. Click the actions menu (three stacked dots) and select **Relocate application**.
4. In the **Relocate application** modal dialog:
   - Verify **Target cluster** indicates `local-cluster`.
   - Verify **Relocate readiness** displays **Ready**.
5. Click Initiate.
6. In the **Disaster Recovery** table, click the status link in the **Data protection** column (such as **Relocating** or **Action needed**) to open the activity drawer.

   The workflow will progress to step 2 (**Clean up**) with the alert:

   `Clean up application resources on the primary cluster ocp2 to start relocating.`

   Note For Discovered applications, ODR pauses at the **Clean up** step waiting for the VolumeReplicationGroup (VRG) on `ocp2` to transition from primary to secondary. Deleting the active pods and PVCs on `ocp2` releases the Ceph RBD volume locks so replication can demote.

7. Switch terminal context to **`ocp2`** and delete the active application resources:

```
export NS=filebrowser
oc -n $NS delete all -l app=filebrowser
oc -n $NS delete cm -l app=filebrowser
oc -n $NS delete pvc -l app=filebrowser
```

8. Return to the `local-cluster` Web Console activity drawer and monitor the remaining automated phases:
   - **Syncing** — Syncs the final storage delta snapshot from `ocp2` to `local-cluster`.

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- **Restoring** — Restores Kubernetes resources (via OADP) and attaches the replicated PVCs on `local-cluster`.

9. Verify that the application pods and PVCs are running on `local-cluster`:

```
oc get pods,pvc -n filebrowser -w
```

10. Confirm DR status from the CLI on `local-cluster`:

```
oc get drpc -n openshift-dr-ops
```

The output should confirm `DESIREDSTATE: Relocate` and `CURRENTSTATE: Relocated`:

```
NAME          AGE   PREFERREDCLUSTER   FAILOVERCLUSTER   DESIREDSTATE   CURRENTSTATE
filebrowser   92m   local-cluster      ocp2              Relocate       Relocated
```

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